



Enhanced power demand forecasting for Bangladesh: using feature engineering associated with environmental and economic impact

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Abstract

Forecasting power demand is crucial for developing countries like Bangladesh for various reasons including resource planning due to limited resources. Limited research was found on short-term power demand forecasting of Bangladesh. In this study, a preprocessing pipeline is proposed to generate powerful features including hourly demand, weather and economic data to generate both short- and medium-term load forecasting. Our method achieved the lowest 2.3% MAPE on PGCB dataset in forecasting energy loads for January and February 2024. The efficacy of the generated features, produced from pre-processing pipeline, was validated by utilising 2 machine-learning models including FB-Prophet and LSTM.

Keywords MTLF · LSTM · Random forest · FB-prophet · Feature engineering · Economic effect · Environmental effect · Day peak · Evening peak

1 Introduction

Bangladesh is one of the fastest growing developing countries of the world with a population of around 169.4 million and GDP of 460.2 billion USD (2022 year) according to the World Bank. For this size of population, the total Power generation capacity of Bangladesh is 22482 MW which is not sufficient to manage the overall electricity demand. In fiscal year 2021–2022, the total energy generation was 85607 Mwh which was 6.45% higher than the previous year. As a developing populated country, Bangladesh's power consumption has a correlation with economic growth which make Bangladesh, an energy dependent country (Masuduzzaman 2012). Electricity demand is increasing with the pace of economic growth of Bangladesh. In order to mitigate this demand with sustainable and renewable energy mix, both

short and long-term load forecasting are necessary (Directorates of System Planning et al. 2022).

Bangladesh faces significant vulnerability to climate change, ranking 159th globally in climate vulnerability. According to UNDP commitments, Bangladesh aims to reduce conditional carbon emissions from 89.47 MtCO₂e to 36 MtCO₂e by 2030. With 97% of its energy sourced from nonrenewable sources (Uddin et al. 2019), the Sustainable and Renewable Energy Development Authority (SREDA) was established to ensure that renewable energy contributes 10% to the total energy supply. Overestimating energy demand could lead to economic losses and exacerbate carbon emissions, impacting both the economy and the environment. Load shedding is a common occurrence in Bangladesh, particularly during summer, affecting rural living standards and economic growth. Consequently, load forecasting is a top priority for energy regulatory authorities.

Accurate load forecasts are essential for effective planning, expansion, and daily management of power systems across the generation, distribution, and planning sectors. Without precise load forecast data, subsequent phases of operation and control is becoming challenging, hindering operational efficiency and security (Rahman et al. 2019). Overestimating load can lead to unnecessary idle capacity and financial losses, while underestimating may result in outages with severe economic and societal consequences including energy blackouts. Therefore, developing a fore-

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casting model for electricity consumption with high precision is crucial to avoid costly errors (Kavaklioglu et al. 2009; Dinakin and Oluseyi 2018).

Load forecasting involves accurately predicting the geographical locations and magnitudes of electric load over various time periods, typically by hours, throughout the planning horizon. It encompasses projections of system energy, peak system load, and values on hourly, daily, weekly, and monthly scales (Gross and Galiana 1987). Load forecasting is categorized based on the length of the planning horizon: short-term load forecasting (STLF) for up to one day, medium-term load forecasting (MTLF) spanning one day to one year, and long-term load forecasting (LTLF) covering periods of 1 to 10 years (Kim et al. 1995; Alfares and Nazeeruddin 2002). In short-term load forecasting, seasonal periodicity is a significant parameter and historical demands, such as demands based on hourly, daily, weekly, and monthly scales along with weather conditions and holiday dates, play crucial roles in the forecasting process (Devison 2016).

Limited research has been done to forecast Bangladesh's electrical power demand. In Power System Master Plan 2016 prepared by Power Division Board, Bangladesh and JICA, long term load forecasting has been done by GDP elasticity method which is not compatible for predicting short term load. In Piyal et al. Piyal et al. (2023), LSTM model gives better accuracy for short term load forecasting of Bangladesh compared to SARIMA and FB-Prophet, but they validate their prediction with only training data which does not reflect model's generalisation capacity.

In this experiment, we preprocess our original dataset and incorporate economic (GDP, Population) and environmental (Temperature, Humidity) variables relevant to energy demand. Additional features were generated using rolling operations to compute statistical values (Mean, Sum, Max) from previous days. The target variable is the next hour demand, enabling future value prediction. This enables generation of load forecasting for both short term (1 hour to 1 day) and medium term (1 month to 2 years). The dataset is divided into training, validation, and testing sets. The training set builds the model, while the validation set assesses model performance. In the testing phase, features are derived from the validation set for prediction, and accuracy is evaluated against actual results over a 45-day period. Contribution of this study can be summarized as below -

- 1) Outliers were eliminated using the IQR approach to improve the Bangladeshi hourly power consumption dataset quality for machine learning.
- 2) Several timestamp variables (year, month, day, hour, and week number) were extracted and examined in conjunction with electricity consumption from the date-time data.

- 3) In order to investigate their association with power demand, holidays and peak hours were included as contextual variables.
- 4) To reflect their substantial association with energy demand in Bangladesh, environmental factors like temperature and humidity and economic factors like GDP and population were incorporated.
- 5) New features were created from the demand data to enhance the model's training, including rolling mean, maximum, and standard deviation.
- 6) After thorough pre-processing and the incorporation of pertinent variables, the LSTM, fb-prophet, and random forest models could predict energy consumption for 45 days with greater accuracy and fewer errors.

2 Literature review

To effectively allocate resources, increase grid stability, lower costs, and better integrate renewable energy sources, power demand forecasting is essential for modern energy systems.

Forecasting is used by National Grid ESO (UK) to assist sustainable transitions by preserving grid resilience, cutting costs, and balancing the supply and demand of renewable energy. Demand forecasting allows Denmark to integrate wind energy for 40% of its power without experiencing volatility. As demonstrated by California's initiatives that reduced peak energy use by 8-10%, forecast-driven demand response systems optimize energy use while lowering operating costs. To avoid overbuilding, long-term projections direct infrastructure investments, and India intends investments in renewable energy to reach its 2030 objective of 450 GW (Williams and Short 2020).

Energy consumption analysis extensively uses various forecasting approaches, including time series, ARIMA, exponential smoothing, multiple regression, and computational intelligence methods, including neural networks, support vector machines, and genetic algorithms (Bunnoon 2011). This study uses artificial neural networks (ANNs) to create mid-term energy load forecasting (MTLF) models to anticipate energy demand two and three years in advance. Although it highlights ANN's potential to lower prediction errors, it ignores outside influences like shifting economic conditions or legislation. Broader applications are limited by the concentration on two- and three-year periods, and it is still unknown how resilient the models are against outliers or missing data (Bunnoon 2011).

ARIMA statistical technique reduced the daily load curve inaccuracy from 2.68% to 2.26% by examining Mongolia's energy consumption with temperature and humidity

data. Using deterministic and stochastic simulations, a statistical model for long-term global energy demand forecasting (1900–2017) investigated the connections between GDP, population, and energy consumption (Rusina et al. 2022). The model emphasizes electrification and renewable energy, pointing out a discrepancy between past trends and zero-carbon ambitions. The necessity for demand-side interventions and policy changes is highlighted by limitations such as a top-down approach, extrapolation hazards, and policy dependence (Rusina et al. 2022).

Turkey's power usage was predicted using Artificial Neural Networks (ANNs), especially a multi-layer perceptron with backpropagation (Kavaklioglu et al. 2009), which took into account economic factors such as GDP, population, imports, and exports. Significant forecast errors occurred during economic volatility due to the Turkish economic crisis 2001. Without systematic optimization, the model relied on trial-and-error approaches for architecture and parameter selection, had problems with extrapolation, and had lower accuracy in time-dependent predictions (Kavaklioglu et al. 2009).

In Prophet-Based Medium and Long-Term Electricity Load Forecasting Research, the Prophet algorithm is employed to forecast electricity load considering variables like seasonality and holidays. Performance evaluation yielded MAPE and RMSE values of 7.43% and 60376.80, respectively, compared to LSTM's 19.18% MAPE and 139643.44 RMSE. The model has some drawbacks, such as not accounting for outside variables like weather, energy legislation, and economic activity; it is also not scalable for larger applications (Long et al. 2022).

Using machine learning techniques, specifically Random Forests (RF) and Artificial Neural Networks (ANN), this study predicts the weekly occurrence of dengue in Colombia (Zhao et al. 2020). While environmental data (temperature, precipitation, and vegetation indices) were resampled into weekly values from daily or 8/16-day intervals, annual sociodemographic indicators (GDP, population, and education coverage) were disseminated weekly. For precise weekly dengue forecasts, the models used variables with varying temporal granularities to capture both short-term environmental changes and long-term socio-economic trends (Zhao et al. 2020).

In Bangladesh, load forecasting has not been studied well. The MAE and RMSE of an LSTM-based study conducted in Chattogram City were 92.448 and 127.682, respectively (Islam et al. 2020). The study's shortcomings include its reliance on scaled historical data, single-region focus, lack of environmental characteristics, manual hyperparameter tuning, and limited model comparisons. Selecting hyperparameters by hand takes much time and might not get the best results. Predictions exhibit minor variations from

actual values because reliance on historical data ignores outside influences such as socioeconomic effects or changes in policy. The model's accuracy is further limited by the absence of environmental parameters (Islam et al. 2020).

In Bangladesh, most forecasting studies are regional, such as the one on Swandip Island, where load increase prediction is based on data from comparable regions, which may need to be more accurate (Mahmud 2011). The methodology's shortcomings include its limited breadth, lack of real-time data, and dependence on linear regression (Mahmud 2011). It also simplifies complex relationships with weighting factors and misses finer load dependencies. Alavi et al. (202) used TDNN for future lockdown scenarios and NARNET to predict power usage during lockdowns in their analysis of PGCB hourly demand data (2019–2020) to investigate the impact of COVID-19 on electricity consumption.

With four years of data, Sadi et al. (2010) used LSTM to produce a MAPE of 3.46, solving the vanishing gradient problem and capturing long-term time-series connections. After preprocessing to eliminate seasonality and trends, the model was trained using hourly load data for a few chosen months. MAE, RMSE, and MPE were used to assess the model's performance. But because it only uses historical load data and ignores outside influences like economic and environmental considerations, it is less useful for abrupt shifts in demand, such as during the COVID-19 pandemic (Rahman et al. 2019). The limited validation of the study concentrates on particular months rather than predictions for the entire year. Furthermore, a high relationship between load and temperature was found using temperature sensitivity forecasting.

3 Methodology

In our experiment, we employ feature engineering and various normalization techniques like Min-Max (García et al. 2015), Max Absolute Scaler (Képešiová et al. 2022), and Standard Scaler (Ferreira et al. 2019) for data processing, along with the Interquartile Range method (IQR) (Wan et al. 2014) for outlier detection. We utilize LSTM, Prophet and Random forest models for energy demand forecasting.

3.1 Pre-processing

3.1.1 Min max scaler pre-processing

Min-Max scaling normalizes numerical data to [0, 1] range, ensuring uniform feature contribution to model training (García et al. 2015), as shown in Eq. 1.

$$X_{out} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

where, X_{out} , X_{min} and X_{max} represents the output, minimum and maximum value of the feature.

3.1.2 Max absolute scaler pre-processing

MaxAbsScaler scales features to have a maximum absolute value of 1.0 (Képešiová et al. 2022), as shown in Eq. 2.

$$X_{out} = \frac{X_n}{X_m} \quad (2)$$

Where, X_n and X_m are the observation and maximum value of the features, respectively.

3.1.3 Standard scaler pre-processing

Standard scaling subtracts each observation by its mean and divides by its variance, normalizing values based on observations' means and variances (Hoaglin et al. 2000; Borg and Groenen 2005), as shown in Eq. 3.

$$X_{out} = \frac{X - \mu}{\sigma} \quad (3)$$

where, X , μ and σ are the actual observation of the features, mean of the features and variance of features, respectively.

3.1.4 Interquartile range (IQR)

The Interquartile Range (IQR) robustly detects outliers by computing the difference between the third (Q3) and first (Q1) quartiles, minimizing outlier influence while indicating dataset dispersion and distribution around the median (Wan et al. 2014).

3.2 Machine-learning model

3.2.1 Long short-term memory (LSTM)

LSTM, a specialized recurrent neural network, adeptly handles long time series data, employing memory cells and gates with sigmoid functions for data flow regulation. Its enhanced architecture enables seamless processing of sequential data (Sunny et al. 2020).

LSTM utilizes gates, including the update, forget, and output gates, represented by Eqs. 4, 5, and 6 respectively.

$$i_t = \sigma(W_{ih}x_t + W_{ix}h_{t-1} + b_i) \quad (4)$$

$$f_t = \sigma(W_{fh}x_t + W_{fx}h_{t-1} + b_f) \quad (5)$$

$$o_t = \sigma(W_{oh}x_t + W_{ox}h_{t-1} + b_o) \quad (6)$$

In the given context, σ , W , $h_{(t-1)}$, and b_i represent the sigmoid activation function, weight matrix, hidden state,

and bias term, respectively. They collectively determine how input information passes through to the cell state, with $h_{(t-1)}$ denoting the previous hidden state.

3.2.2 Prophet

Prophet, developed by Facebook, is a versatile forecasting tool for Python and R, ideal for data with hourly, daily, or weekly observations over several months. It identifies trend changes and handles nonlinear growth curves, incorporating regular and irregular holidays in forecasts (Rafferty 2021; Satrio et al. 2021; Harvey and Peters 1990). Its additive regression model includes trends, seasonality, and holidays, estimating seasonality with partial Fourier sums. The equation for the model is,

$$f(t) = tr(t) + se(t) + hl(t) + e \quad (7)$$

In Eq. 7, $tr(t)$ models non-periodic changes, $se(t)$ represents seasonal periodicity, $hl(t)$ captures holiday impacts, and e accommodates unique changes in the time series.

In prophet model two types of growth model can be used which are linear growth (McGee and Carleton 1970) or logistic growth model (Jha and Pande 2021) for forecasting. For our experimental dataset we used linear growth model. The mechanism of linear and logistic growth model are below.

$$f(x) = \frac{m}{1 + e^{-g(x-x_0)}} \quad (8)$$

$$f(x) = a + bt + \epsilon_t \quad (9)$$

Equation 8 illustrates logistic growth, with m as the maximum value, g as the growth rate, and x_0 as the sigmoid point. Equation 9 depicts linear growth, with a as the initial value, b as the rate of change, t as time, and ϵ_t as the error term.

3.2.3 Random forest regression (RF-R)

RF-R is an ensemble learning approach that mixes numerous decision trees to increase forecast accuracy while minimizing over-fitting (Segal 2004; Dietterich 1995). It generates many trees from distinct subsets of data and averages their predictions to provide a more robust result. RF-R decreases variation by averaging results, making it suitable for complicated and nonlinear interactions. Equation 10 gives the mathematical representation of RF-R.

$$output = \frac{1}{N} \sum_{i=1}^N T_i(x) \quad (10)$$

where, $T_i(x)$ and N represents the prediction from the $i - th$ decision tree, and the number of trees, respectively.

Table 1 Statistical description of variables

	Demand	Temperature	Humidity	GDP	Population
Mean	8363.62	25.68	78.82	2128.88	1.7E+08
Std	2341.47	4.17	12.67	445.25	4.8E+06
Min	3578	12.66	33.50	1236	1.6E+08
Q1	6597	23.04	71.38	1816	1.6E+08
Q2	8014	27.64	82.62	2122	1.7E+08
Q3	9927	28.57	89.00	2458	1.7E+08
Max	15128	32.90	95.56	2793	1.7E+08
Skewness	0.48	-0.98	-1.04	-0.26	-0.07
Kurtosis	-0.40	-0.13	0.42	-0.71	-1.20

4 Dataset description and experimental setup

4.1 Dataset description

The study utilized a dataset from the Power Grid Company of Bangladesh Limited (PGCB), covering the period from '2015-04-17 17:30:00' to '2023-12-31 23:00:00', providing electricity demand data in Megawatt (MW) on an hourly basis, totaling 78,906 rows. Employing the Inter Quartile Range (IQR) method (Wan et al. 2014), 344 outlier points were identified and removed, constituting 0.43% of the dataset. Following this adjustment, the demand range observed across all years ranged from 3,500 to 15,500 MW.

We combine the raw demand dataset with economic (GDP, population) and environmental (temperature, humidity) factors. There is a frequency mismatch since power consumption is hourly, whereas environmental and economic variables are daily averages and annual totals, respectively. The integrity of the statistics may be compromised if GDP and population are converted to lower frequencies. Daily temperature averages, which change very little from the daily minimum and maximum values, are utilized for environmental variables. As in earlier research, we maintain constant annual economic factors and set the temperature for the entire day to correspond with hourly consumption. Zhao et al. (2020); Frikha et al. (2024); Bilal et al. (2022).

After integrating the economic and environmental factors, we applied the IQR approach to the dataset, and Table 1 summarizes the results. Every variable's average, minimum, maximum, standard deviation and quartiles are shown in the table. It demonstrates that humidity has high kurtosis, which indicates heavier tails and electricity usage has low kurtosis, which means lighter tails are needed for outlier elimination. The population's skewness is about 0, indicating the data is symmetrical.

Figure 1 represents the relationship of feature variables with target variable where GDP and population have higher relation and temperature have higher than humidity.

Figures 2-a to -c depict the density distribution for the years [2015-2017], [2018-2020], and [2021-2023], respectively, while Fig. 2-d illustrates the increasing mean demand of Bangladesh annually. Bangladesh's electrical demand density rose throughout time, as seen in Figs. 2-a to -c: 6000-8000 (2015-2017), 6500-9000 (2018-2020), and 7500-10000 (2021-2023). However, Fig. 2-d shows an exception for 2019-2020 because of the COVID-19 pandemic, which reflects lower electricity usage in government offices and institutions. This tendency is supported by yearly average demand.

The dataset includes hourly demand data from the PGCB, identifying three peak timings: Normal peak, Day Peak, and Evening Peak. Day Peak spans from 8:00 AM to 11:00 AM, while Evening Peak covers 5:00 PM to 8:00 PM. Figure 3-a displays the average demand by month for these peak tim-

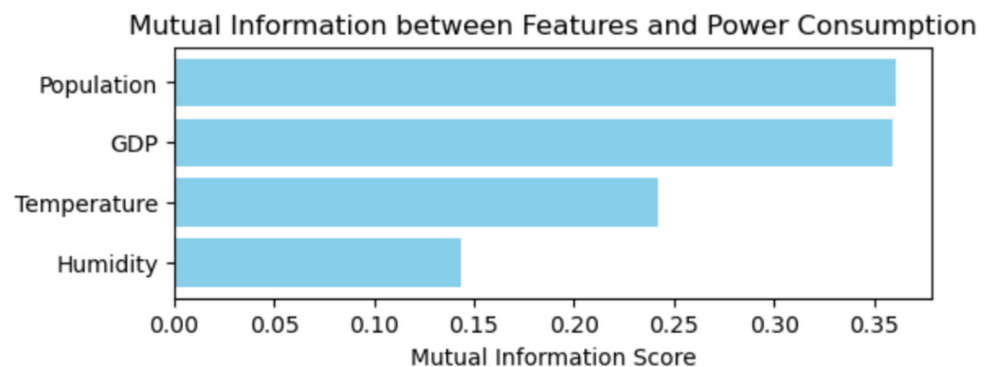
Fig. 1 Mutual information with feature variables

Fig. 2 Density Distribution of Electricity Demand for duration: (a) 2015–2017, (b) 2018–2020, and (c) 2021–2023; finally, (d) Mean Electricity Demand by Year

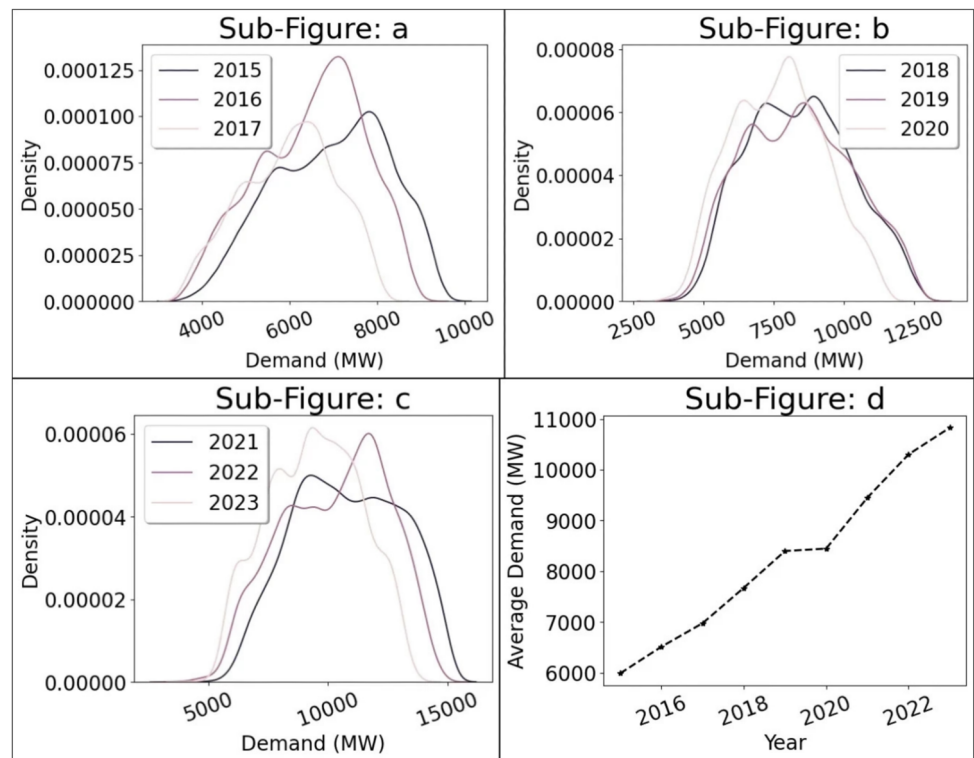
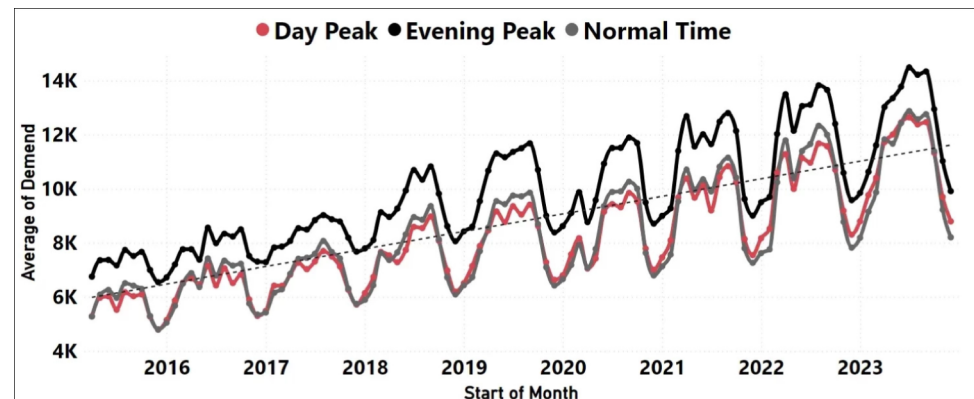
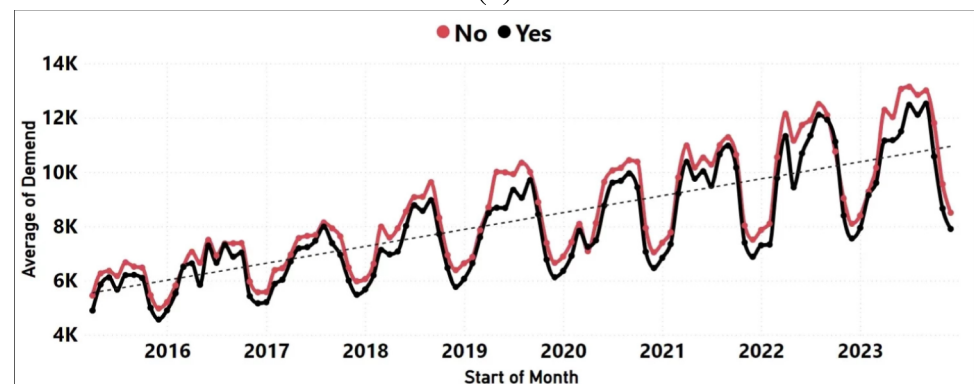


Fig. 3 (a) Day Peak Average Demand Month-wise, and (b) Average Demand Month-wise for Holiday



(a)



(b)

ings, with Evening Peak showing the highest demand. So highest demand of evening peak shows a huge use of electricity in household activities and also in the industries. From 5:00 PM to 8:00 PM people use highest amount of electricity than other time of a particular day. Additionally, a holiday feature column was added, illustrated in Fig. 3-b, indicating lower demand on holidays due to office and commercial closures. Although the difference is very low but in the middle and end of every year the difference is higher than other times. Also it shows an increment trend by increasing of year which identify the more uses of energy in Bangladesh due to digitalization.

4.2 Data analysis

Analysing of the dataset yields significant impacts on the economy and specific times of a day for electricity usage in Bangladesh. Particularly during the COVID-19 pandemic, electricity consumption exhibited more pronounced changes than normal.

We analyzed an hourly time series dataset containing peak hour information. Figure 4 shows an increasing trend in average electricity demand during day and evening peaks, with evening peaks significantly higher. However, demand was stagnant in 2019 and 2020 due to the COVID-19 pandemic, a trend observed globally. This indicates future pandemics might cause similar demand patterns. Blue and black dotted lines represent day and evening peak ranges, respectively.

We included a government holiday variable in our dataset. Figure 5 shows that, during day peaks, working days consume more electricity than holidays due to active offices and commercial zones. In evening peak range, the consumption difference is small for both holiday and working day. Notably, during COVID-19 (March-June 2020), holidays consumed

more electricity than working days due to lockdowns and closed organizations.

4.3 Environment setup

In our research, Python, a versatile language, was employed, well-suited for data science (Sanner 1999). Utilizing pandas and Scikit Learn libraries (McKinney 2012; Fabian 2011), we manipulated data and conducted machine learning tasks. pandas facilitated data handling and analysis, while Scikit Learn supported machine learning and data pre-processing. We employed the train-test split process for validation (Vabalas et al. 2019).

4.4 Performance measurement metrics

In our experiment, we applied Fb-Prophet and LSTM machine-learning models to predict outcomes and utilized performance metrics for evaluation and comparison of model performance.

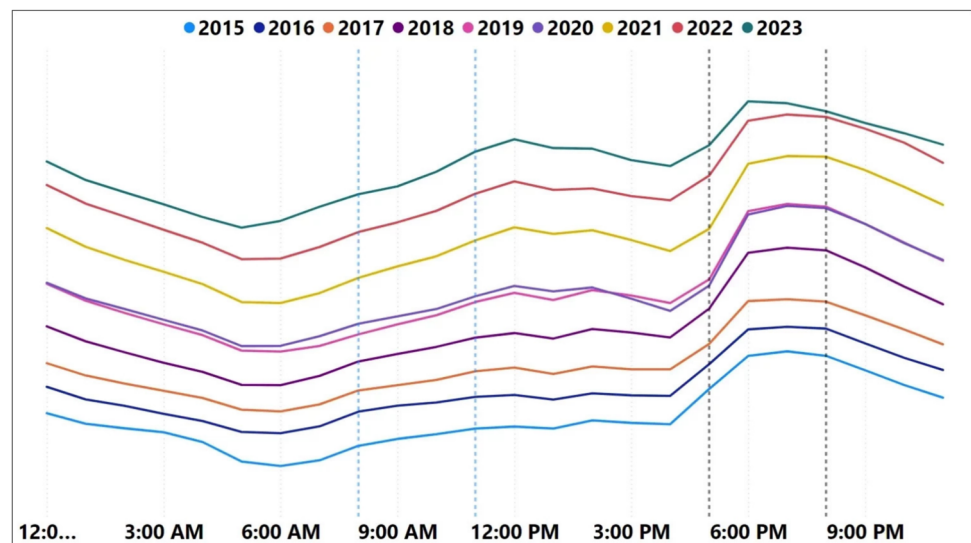
4.4.1 Root mean square error (RMSE)

RMSE, a performance measure for regression models, involves squaring the differences between actual and predicted values, calculating their mean, and then taking the square root of the result (Chai and Draxler 2014), as shown in Eq. 11.

$$RMSE = \sqrt{\frac{1}{N} \sum_{k=1}^N (y_k - \hat{y}_k)^2} \quad (11)$$

where, N , y_k and $\hat{y}_k$ are the total observation, actual value and predict value, respectively.

Fig. 4 Year wise Average Demand for Peak Time



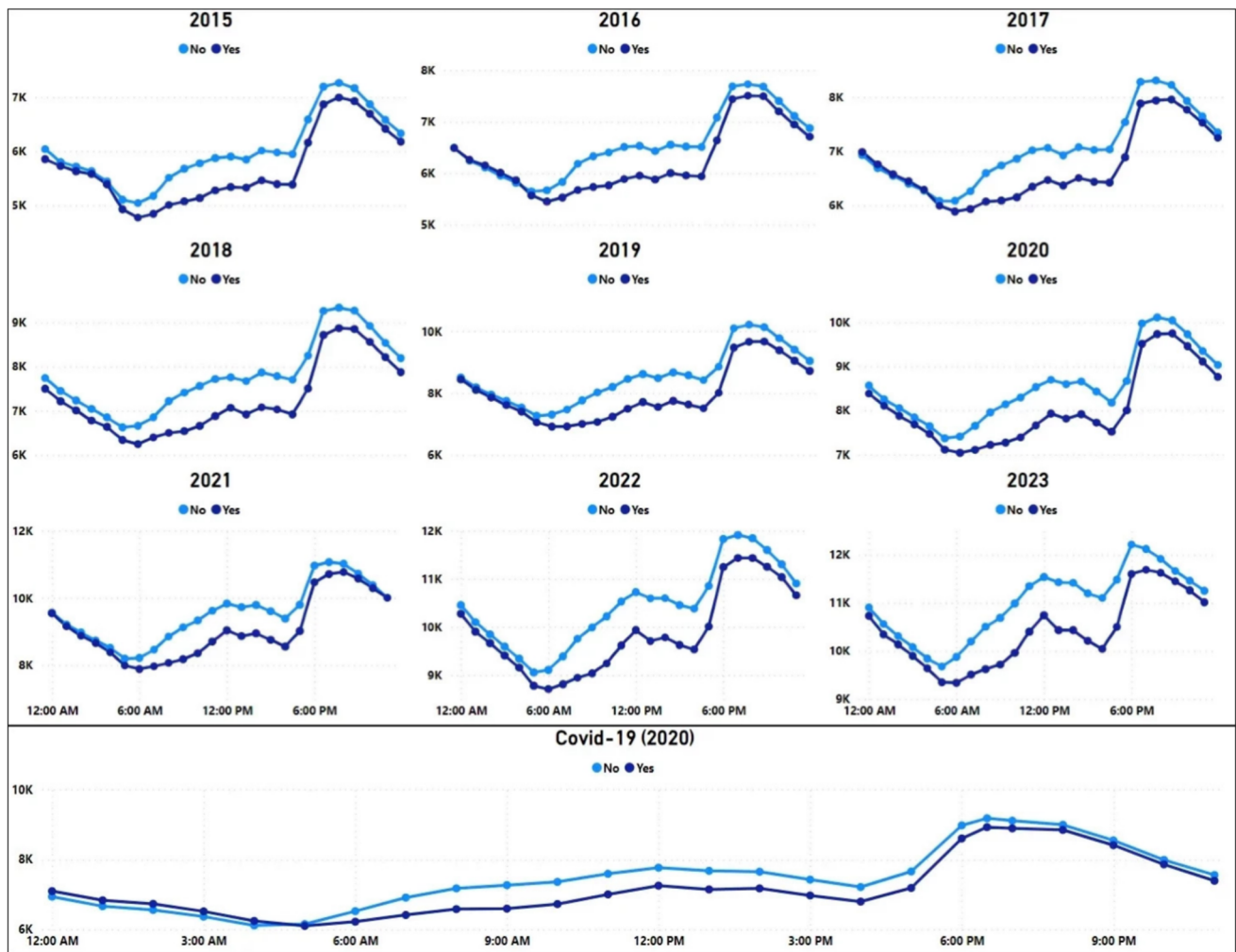


Fig. 5 Holiday wise Average Demand Analysis

4.4.2 Mean Absolute Error (MAE)

MAE, an accuracy indicator for predictions, measures the average difference between actual and predicted values (Chai and Draxler 2014), as illustrated in Eq. 12.

$$MAE = \frac{1}{n} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (12)$$

where, N , y_i and $\hat{y}_i$ is the total observation, actual value and predict value, respectively.

4.4.3 Mean absolute percentage error (MAPE)

MAPE assesses forecasting accuracy by averaging absolute percentage mistakes of each item, comparing anticipated to

actual quantities (De Myttenaere et al. 2016), as depicted in Eq. 13.

$$MAPE = \frac{1}{N} \sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right| \quad (13)$$

The difference between the actual value A_t and the forecasted value F_t is divided by A_t .

4.4.4 The coefficient of determination, or R-squared (R^2)

R^2 indicates the goodness of fit of a model to its data, crucial for forecasting future events, by measuring how well the regression line fits the actual data (Cameron and Windmeijer 1997), as shown in Eq. 14.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (14)$$

where, y_i , $\hat{y}_i$ and $\bar{y}$ are the actual value, predict value and mean values, respectively.

5 Process flow architecture

The overall process of our system is divided into 4 main parts. Those are:

- 1) Incorporate additional Feature.
- 2) Feature Engineering and Column Creation.
- 3) Train Validation Process.
- 4) Data Forecasting Process.

All parts are connected with each other using logical manner. The description of those 4 parts are below and Fig. 6 represents the overall flow of our experiment:

5.1 Additional feature

In our research paper, we conducted time series analysis on the energy demand dataset of Bangladesh. Alongside this,

we incorporated supplementary information relevant to this dataset. This additional data falls into three distinct categories, such as:

- Economic: GDP and total population of Bangladesh.
- Feature Engineering and Column Creation.
- Environment: Day wise Temperature and Humidity.
- Date-time information: Government holiday of Bangladesh and Day Peak information.

5.2 Feature engineering and column creation

We engineered date-related features from our periodically incremental dataset, capturing hourly, daily, and monthly patterns. Additionally, recognizing the correlation between current and past energy demand, we computed rolling statistics (Mean, Max, and Sum) to retain the historical demand characteristics. The feature engineering process, which categorizes variables into three groups to construct machine learning models, are as follows:

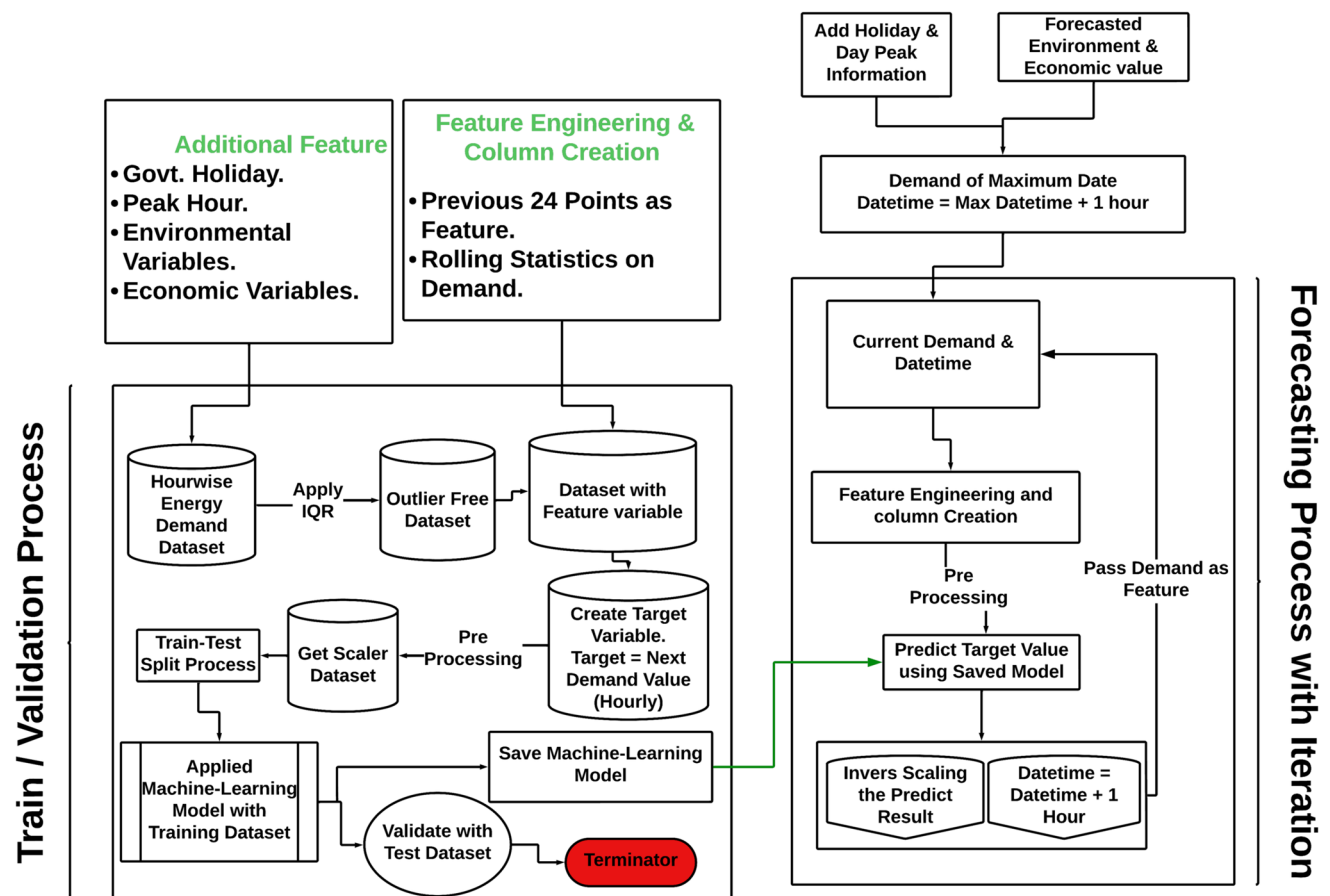


Fig. 6 Overall process flow architecture of our experiment

5.2.1 Date feature

Our time series dataset reveals recurring demand patterns within daily, monthly, and quarterly intervals across multiple years. To capture these patterns effectively, we include features such as day number of year, day number of month, month number of year, quarter number of year, year value, and hour number from date-time information. Additionally, to account for weekly fluctuations in demand, we incorporate features like day number of week and week number of year into our Machine Learning model.

5.2.2 Previous 24 points As feature

Lagging in time-series forecasting, known as the 'lagging problem,' causes predictive values to lag behind truth values, particularly noticeable in recurrent network models like LSTM for financial forecasting (Surakhi 2021). Optimal lag selection is crucial for accuracy, as it improves model performance by capturing relevant historical information (Ribeiro 2011). In our experiment with hourly data, we address this by incorporating lagged features from the previous 24 points into the machine-learning model. Given the distinct day and evening peak demand periods in our dataset, leveraging the lag operation with the preceding 24 data points provides valuable insights for constructing a robust model. Thus, we utilize these lagged data points and their ratios as feature variables to enhance model effectiveness.

5.2.3 Statistical variable using rolling operation

Rolling is a valuable technique for time series data, involving the establishment of a rolling window of a specific size

for computations on the data within it, which progresses through the dataset (Zivot et al. 2003; Kumar and Jain 2010). Rolling origin assessment entails moving the forecasting origin forward by a defined number of observations, generating predictions from each origin (Tashman 2000). This approach yields multiple forecast errors, offering insights into model performance. Computation begins after the entire window is present in the data, with the first aggregation occurring on the N-th row if the window size is N.

In our experiment, we utilize rolling operations (Mean, Sum, Maximum, and Standard Deviation) on demand variables at various previous data points (2, 3, 5, 7, 9, 11, 13, 15, 17, 19, 20, 21, 22) to create feature variables. Given the hourly dataset's consistent patterns for peak and off-peak hours, leveraging these rolling operations on previous data points provides a clearer understanding of historical patterns, aiding in machine learning model training.

5.3 Train validation process

- First, we augment the original dataset by adding additional features such as Day peak, Environmental, and Economic variables.
- Subsequently, we apply the IQR method in demand variable to identify outliers. Out of 78,906 rows in our dataset, 344 are flagged as outliers (0.43% are outlier which is very less amount) and removed.
- Following outlier removal, we implement the *Feature Engineering and Column Creation* mechanism to generate feature variables from the modified dataset.
- Then, we create the target variable by shifting the next demand value from the existing energy demand dataset,

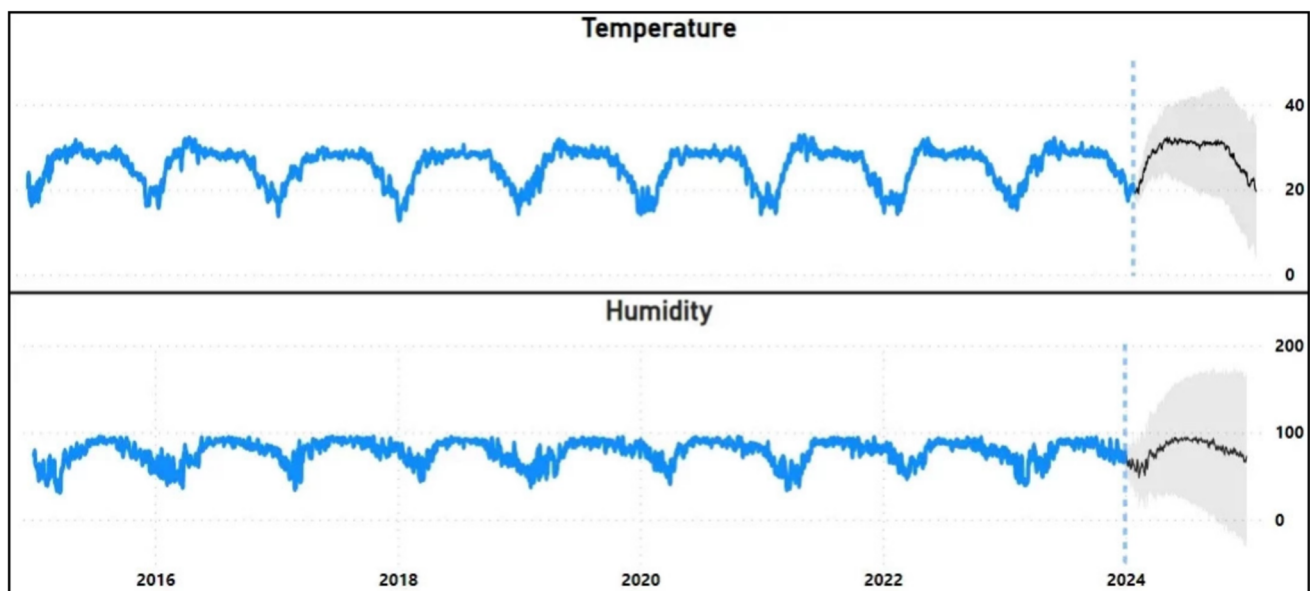


Fig. 7 2024 Temperature and Humidity (Forecasted) for Machine-Learning model variable

making it represent the energy demand for the subsequent hours.

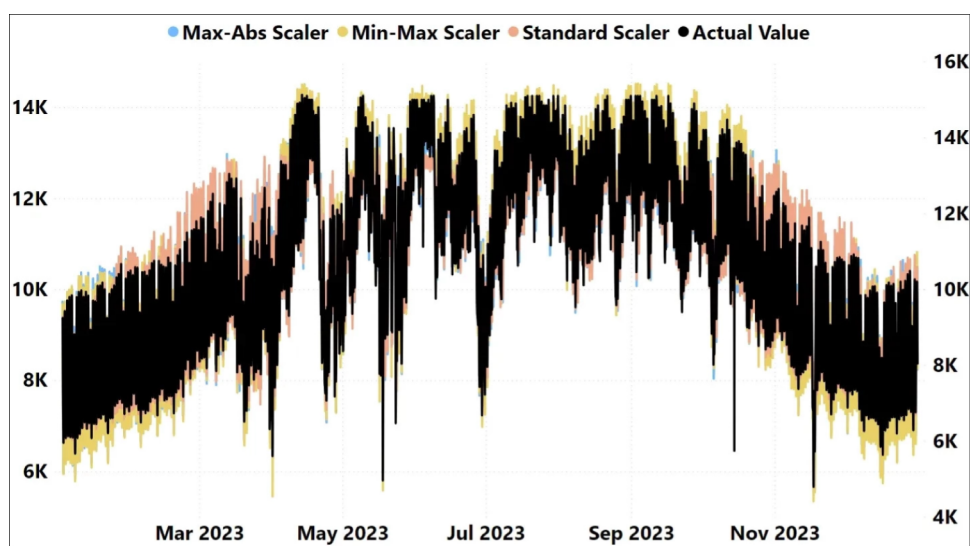
- Data pre-processing is a crucial step in machine learning to streamline model complexity (García et al. 2015; Kotsiantis et al. 2006). Therefore, we employ pre-processing algorithms : Max Absolute, Min-Max, and Standard Scaler methods on the feature-engineered dataset to ensure model suitability and effectiveness.
- Given that the dataset spans 9 years, we divide it into a train-test (validation) split. Data from 2015 to 2022 constitute the training dataset, while data from 2023 serves as the test (validation) dataset. Subsequently, we train our machine-learning model using the training dataset and validate it using the test dataset.

- Additionally, we save our machine-learning model for future forecasting processes.

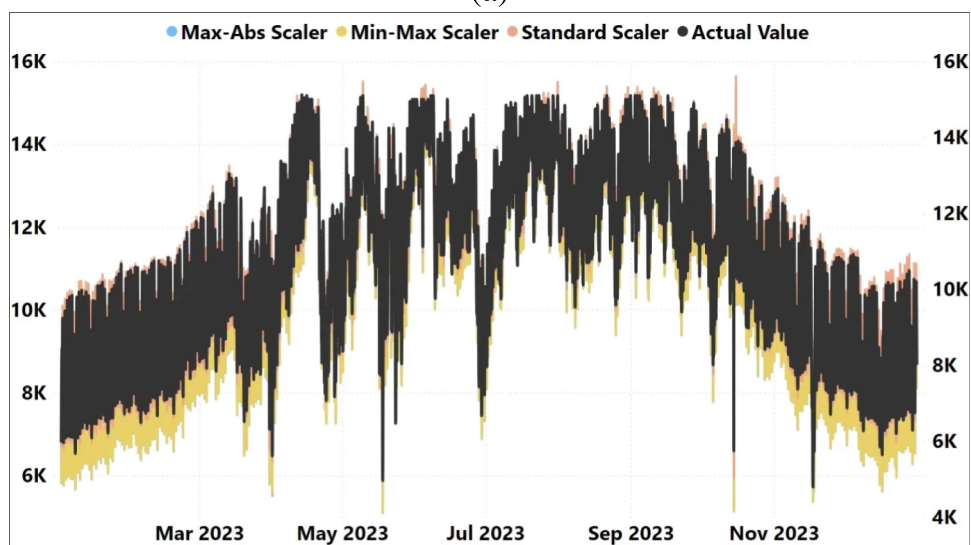
5.4 Forecasting process

In this experiment, we utilize previous demand data and conduct feature engineering to create additional columns for predicting the target variable, energy demand for the next point. Economic and environmental variables such as temperature, humidity, GDP, and total population play crucial roles in energy demand forecasting. However, obtaining future actual values for these variables is impractical. To address this challenge, we forecast temperature and humidity using the Microsoft Power BI tool (Belghith et al. 2023), as depicted

Fig. 8 Actual and Predicted value compare for the Test Dataset using (a) LSTM, (b) FB-Prophet



(a)



(b)

in Fig. 7. Additionally, GDP and total population forecasts for 2024 are sourced from the IMF (International monetary fund 2023), with values of 2646 USD and 174, 701, 211 respectively. The outlined steps detail the power demand forecasting process.

- Initially, we obtain the date-time by adding one hour to the maximum date from the existing dataset and retrieve the demand from the previous day's target value for the current calculated date-time.
- Next, we incorporated additional features and applied feature engineering process to the current calculated date-time and demand.
- We employed data pre-processing techniques to create features, which were then utilized in the train/validation process.
- Subsequently, the feature variables were passed to the saved machine-learning model to predict the target value, which represents the demand for the next hour. Since

scaled values were used for the machine-learning model, reverse engineering was necessary to obtain the original target value.

- Once the target and next calculating date-time were obtained, we proceed with the forecasting approach recursively hour-by-hour forward in time.

6 Simulation and result analysis

In our experiment, we employed LSTM, Fb-Prophet and random forest models to assess results across training, validation, and prediction. Various preprocessing algorithms were used to scale our dataset, producing diverse outcomes. Figure 8-a and -b display actual and predicted demand using LSTM and Fb-Prophet, respectively. The comparison highlights that different preprocessing methods yield varying outcomes for the same time series. Notably, FB-Prophet outperforms LSTM and random forest in our experiment,

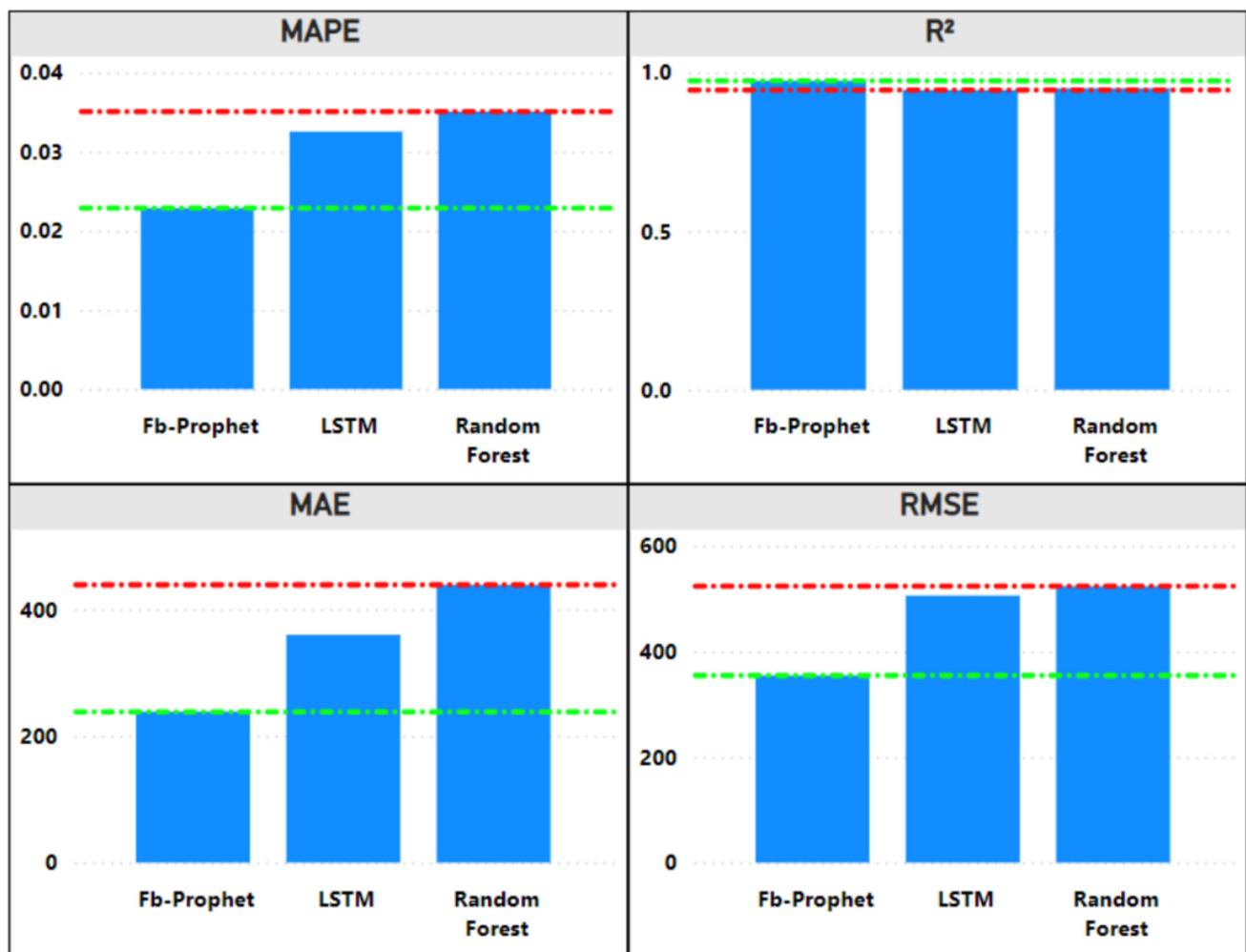


Fig. 9 Result compare of different Machine-Learning model with different evaluation metrics

Table 2 Performance Evaluation Metrics for Random Forest

Data Pre-Processing	Data Split	RMSE	MAE	MAPE	R^2
Max Absolute Scaler	Train	452.94	357.3	4.87%	0.95
	Validation	697.2	593	4.46%	0.925
Min Max Scaler	Train	369.65	294.82	3.4%	0.965
	Validation	437.09	361.43	3.03%	0.963
Standard Scaler	Train	369.65	293.02	3.54%	0.954
	Validation	437.09	361.43	3.03%	0.963

especially during changing demand periods in April to July and in September, providing more accurate predictions aligned with actual data.

Figure 9 compares different evaluation metrics using machine-learning algorithms where evaluation metrics show the average value of different pre-processing techniques. In Fig. 9, fb-prophet model gives better results than others models where the horizontal green line shows the best value and horizontal red line indicates the worst value achieved by the model. The R^2 value shows marginal difference between min and max value ranges.

Tables 2, 3 and 4 summarize the performance evaluation metrics for random forest, LSTM and FB-Prophet, respectively, providing a clear overview of training and test (validation) results for our dataset. Across both time series algorithms, Min-Max Scaler yields superior output compared to the other two preprocessing algorithms. Additionally, we observe that FB-Prophet exhibits a higher Mean Absolute Percentage Error (MAPE) in training than in validation. This suggests that FB-Prophet delivers better results for the validation dataset.

In our experiment, we forecasted results for the year 2024, as shown in Fig. 10. Comparing LSTM with FB-Prophet revealed interesting patterns. LSTM displayed significant irrelevant patterns compared to the original dataset from previous years, especially when employing Standard Scale Pre-Processing, which failed to capture the output pattern from previous years effectively. In contrast, the FB-Prophet algorithm delivered better results overall. However, it showed slightly different patterns, particularly in March to June and the middle of December.

For forecasting, we derived our target from the previous demand value point without using the original values from our dataset. Subsequently, we compared the forecasted demand values with the original values. Figure 11 illustrates the original and predicted values for different data pre-processing methods using FB-Prophet. These predicted values were generated without access to the original dataset. We observed that Max Absolute Scaler yielded superior output compared to other methods. Since these forecasted values were solely based on model output, without knowledge of the original dataset, they tended to be lower than the training and validation parts.

Figure 12 depicts the residual value analysis for day-wise data points employing three different pre-processing methods. Residual values were calculated using Eq. 15 to determine the difference between actual and predicted outputs. The figure indicates that the Max Absolute Scaler method is notably closer to the parallel X-axis dashed line compared to other pre-processing outputs, suggesting it exhibits less difference from the actual values.

$$Residual = actual - predict \quad (15)$$

Table 5 summarize the performance evaluation metrics for forecasted output using FB-Prophet and Max Absolute Scaler gives better result in terms of RMSE, MAE, MAPE and R^2 . Max Absolute Scaler gives 23% and 38% better R^2 ; 6% and 18% better MAPE than Min Max scaler and Standard scaler preprocessing. For the better forecasting output, combination of Max Absolute and fb-prophet can be a good option for the Bangladesh power consumption.

Table 3 Performance Evaluation Metrics for LSTM

Data Pre-Processing	Data Split	RMSE	MAE	MAPE	R^2
Max Absolute Scaler	Train	342.84	233.91	3.08%	0.974
	Validation	572.24	410	3.67%	0.93
Min Max Scaler	Train	245.7	166.32	2.20%	0.987
	Validation	367.79	255.71	2.33%	0.972
Standard Scaler	Train	293.54	199.34	2.58%	0.981
	Validation	577.57	413.48	3.76%	0.931

Table 4 Performance Evaluation Metrics for FB-Prophet

Data Pre-Processing	Data Split	RMSE	MAE	MAPE	R^2
Max Absolute Scaler	Train	278.58	189.22	2.46%	0.983
	Validation	354.03	237.94	2.28%	0.974
Min Max Scaler	Train	278.72	189.5	2.46%	0.984
	Validation	354.4	238.23	2.29%	0.974
Standard Scaler	Train	278.02	188.78	2.46%	0.983
	Validation	355.19	238.51	2.29%	0.974

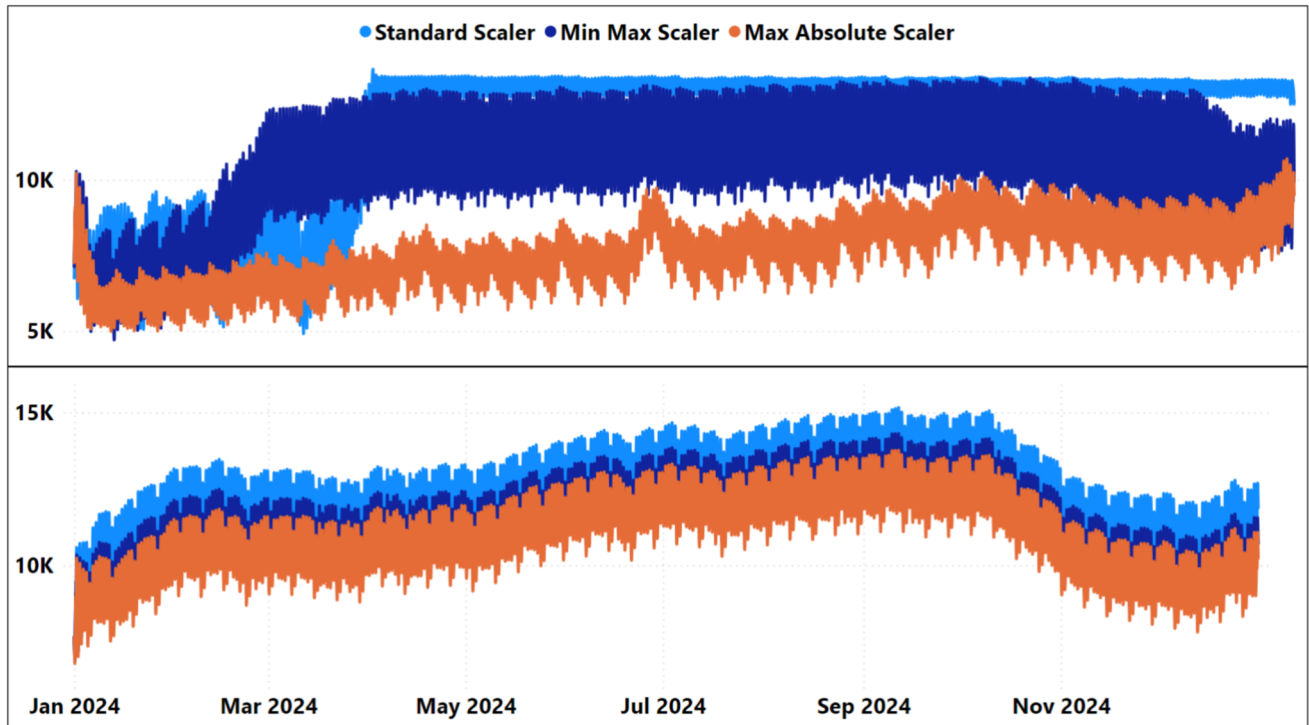
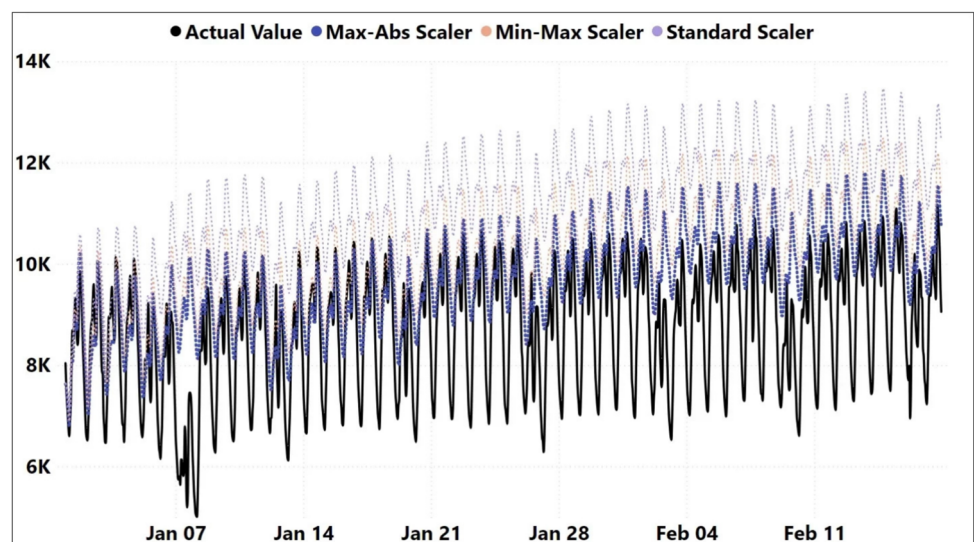
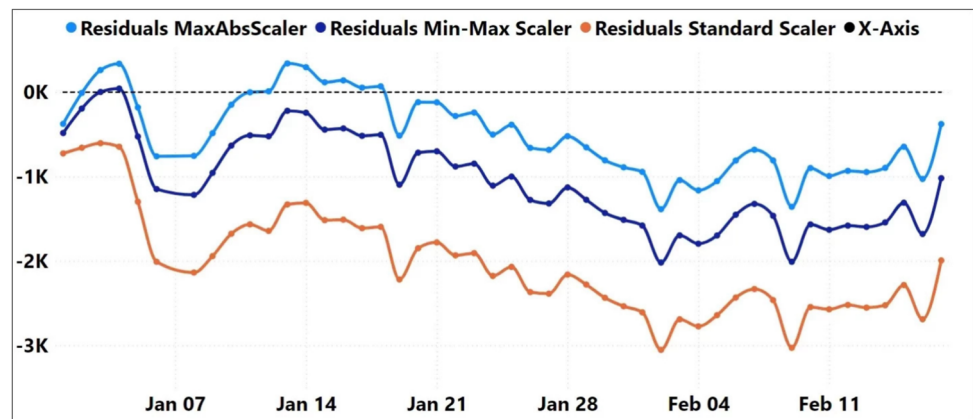
**Fig. 10** Forecasting result for 2024 year using LSTM (top) and FB-Prophet (bottom)**Fig. 11** Forecasting output compare with original output using Fb-Prophet

Fig. 12 Residual Plot Compare with Actual and Forecasted Value using Fb-Prophet



7 Discussion

This study analysed electrical energy demand hourly data during period 2015–2023 and generated hourly demand forecasting. The key findings of this study includes the following: 1) hourly demand forecasting was generated and rolled over the months of January and February 2024, which was the test data split, and achieved MAPE score 2.3%, 2) predicted demand of next hour was used as input demand to the model, in addition to the previous hours with one less, to calculate the following next hour which required to reverse the feature engineering process to generate feature values, 3) the pre-processing pipeline was able to generate forecasting with lower MAPE error with more than one machine learning model (3 models - Random Forest, LSTM and FB-Prophet) indicating effectiveness of generated features, and 4) the use of contextual information, such as environmental data (temperature and humidity) and economic data (GDP and population), was found necessary to achieve better performance.

Incorporation of environmental information was necessary since weather condition is one of the most important factor of electricity demand.

Various environmental and economic factors impact electricity demand in Bangladesh. The country experiences heavy seasonal rainfall, high temperatures, and humidity, following a cyclical pattern throughout the year. Demand rises with increasing temperatures annually, while humidity patterns, albeit following a periodic trend, exhibit a three-month

shift compared to demand. This observation is supported by recent studies (Khan et al. 2019; Istiaque and Khan 2018; Parkpoom et al. 2004). Figure 13-a illustrates the dataset's increasing demand trend over the years, with the trend-line consistently surpassing previous years. In Fig. 13-b, the relationship between demand, temperature, and humidity is depicted, highlighting a strong correlation between temperature and demand, while humidity patterns exhibit a slight deviation from demand trends.

Economic fluctuations in Bangladesh are closely tied to power consumption, with periods of economic growth leading to increased electricity demand, while downturns may result in reduced consumption due to decreased industrial activity and consumer spending (Arshad and Beyer 2023). In Fig. 13-c, the third sub-figure illustrates the correlation between GDP growth, population growth, and electricity demand in Bangladesh, showing a proportional increase in all three factors annually. Given Bangladesh's status as a developing country experiencing population growth, the demand for electricity is on the rise. From analysis, it is observed that the Demand column (independent variable) exhibits strong correlations with economic variables such as GDP (correlation of 0.61) and population (correlation of 0.63). In contrast, for environmental category variables, Temperature shows a correlation of 0.49 with Demand, while Humidity demonstrates a weaker correlation of 0.05, which is lower than that of the economic factors.

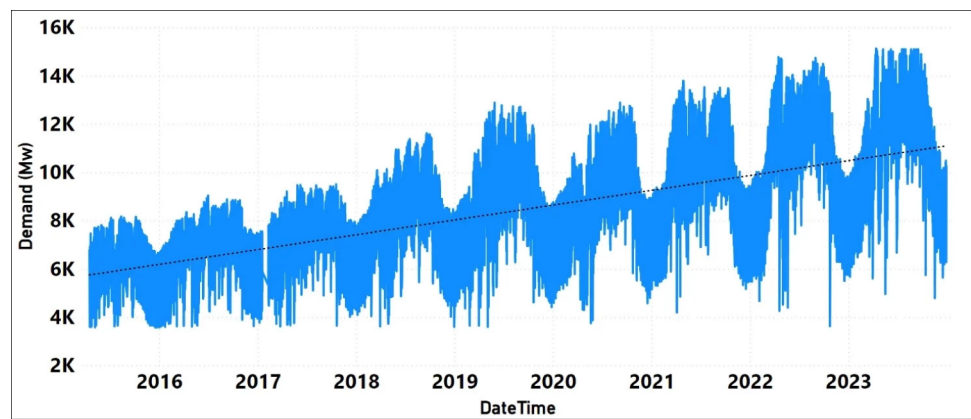
Figure 14 illustrates the correlation between demand growth and economic growth, particularly GDP growth. It's evident that demand growth is closely associated with GDP growth. During the COVID-19 pandemic in 2020, both demand and GDP experienced a decline compared to the previous year. However, in the subsequent years, GDP growth showed an upward trend alongside electricity demand. Interestingly, over the last two years, growth has decreased in Bangladesh.

In our study, we incorporated various economic and environmental factors alongside statistical rolling operations,

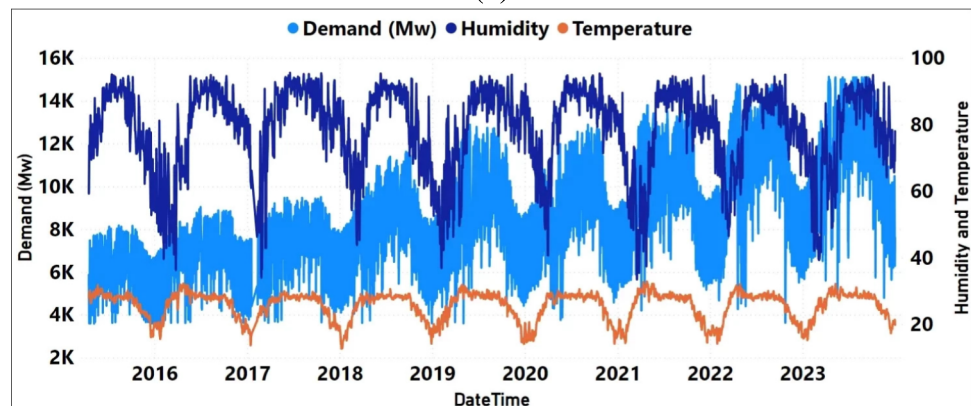
Table 5 Performance Evaluation for demand forecasting using fb-prophet model

Pre-Processing	RMSE	MAE	MAPE	R^2
Max Absolute Scaler	1401.56	1103.5	14.69%	0.65
Min Max Scaler	1868.51	1590.55	20.50%	0.5
Standard Scaler	2790.94	2568.58	32.15%	0.4

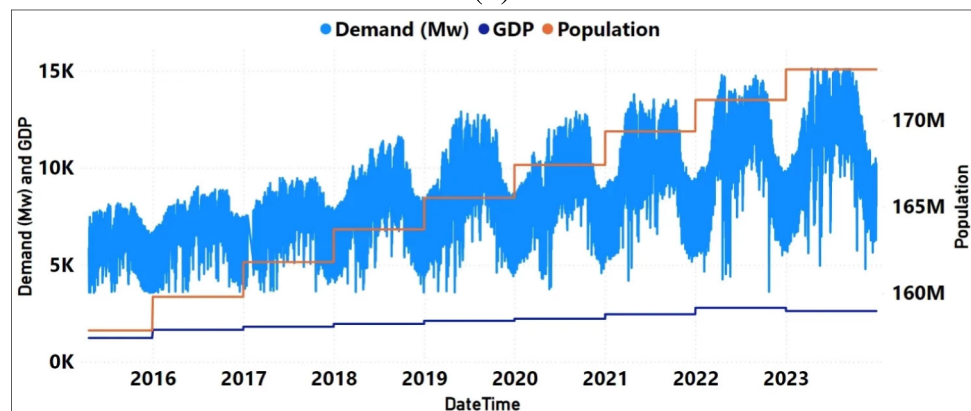
Fig. 13 (a) Hourly Demand Dataset and Year-wise Trend, (b) Demand Data and Environmental Data (Temperature & Humidity) Relationship, and (c) Demand Data and Economic Data (Population & GDP) Relationship



(a)



(b)



(c)

enhancing the complexity of our model. Employing Medium Term Load Forecasting (MTLF) allowed us to forecast 1 month to 2 years, considering the influence of environmental and economic factors on long-term energy demand. Previous research Parkpoom et al. (2004); Arshad and Beyer (2023) underscores the significance of these factors in energy demand prediction. Our dataset's similarity to previous data points accentuates the importance of rolling operations,

contributing to our model's complexity and improved load forecasting performance, as depicted in Fig. 12.

Short-term, as well as, medium-term power demand forecasting studies exist in literature, but similar studies on the dataset used in our study is negligible. Only 2 studies exist, as per our finding till now, but one of them is difficult to compare due to difference in data splitting strategy and volume of data used.

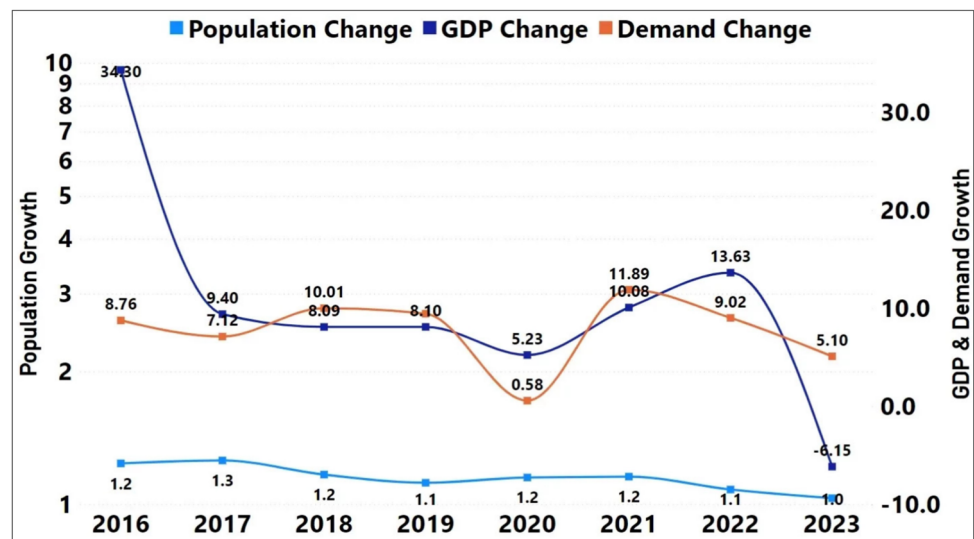
Fig. 14 Economic and Demand Growth Analysis

Table 6 shows performance, metric and data split of studies. Islam et al. (2020) focused on partial dataset, Chattogram City data, from 2015 to 2019, reporting LSTM-generated RMSE of 92.448 and MAE of 127.682 without reference for the assumption that Chattogram consumes only 11% of overall Bangladesh demand. Rahman et al. utilized LSTM but avoided seasonality and trend, yielding MAPEs ranging from 3.5% to 5.06% (Rahman et al. 2019), whereas our approach achieved a lower MAPE range of 2.3% to 2.5%. Piyal et al. (2023) used PGCB dataset consisting of 11-years data from 2003 to 2014 and using LSTM model, MAPE score 0.48% was reported, however, it was found that MAPE formula used in that study was incorrect which is, in fact, the formula of MAE.

Our study utilized a comprehensive dataset covering demand data for Bangladesh from 2015 to 2023, employing LSTM and fb-Prophet algorithms, resulting in a MAPE of 2.3% for demand prediction.

8 Conclusion

In our study, we used time series data for energy demand consumption and added environmental and economic variables to evaluate the socioeconomic and ecological impact of the variables. Our study showed the seasonal effect of peak

hours and holidays on energy consumption demand. These variables have different effects on energy consumption, and our analysis shows the positive impact of GDP and temperature on the power demand. The GDP growth and energy consumption are directly related. We have shown the analysis of the varying usage of power consumption in Bangladesh during the COVID-19 pandemic, which may provide insight to help the policy makers in power management in any future pandemic situation.

We have used machine-learning technique with a proposed pre-processing pipelines to generate powerful features in forecasting hourly energy loads from historical hourly power demands. Contextual information, such as weather data (temperature and humidity) and economic data (population and GDP), were included to enhance the completeness of the feature-set, yielding the lowest MAPE error of 2.3% on the PGCB dataset in forecasting demand for January and February 2024. The strength of the generated features was validated using three machine-learning models including Random Forest, fb-prophet, and LSTM.

As we used the environmental variables (temperature and humidity) to construct our machine-learning models, we had to forecast them proportionately for the demand forecasting. A future direction could be to use only the power demand time-series for future demand forecasting. Using the proposed pre-processing pipeline in another dataset of similar

Table 6 Comparison between power demand forecasting among relevant studies

Author	Data	Performance Metric	Score
Rahman et al. (2019)	PGCB full	MAPE	3.50-5.06%
Islam et al. (2020)	PGCB partial	MAE	127.682
Piyal et al. (2023)	BPDB data	MAPE	0.48%
Our study	PGCB full	MAPE	2.30-2.50%

or different geographical and economic contexts would be an interesting future research direction.

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Data Availability The dataset used in the current research was obtained from Power Grid Bangladesh PLC of hourly generation and load shed (https://erp.pgcb.gov.bd/web/generations/view_generations).

Declarations

Competing Interests statement The authors have no conflict of interest to disclose.

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